

Task 1 Report

Human Rule Formalization of §47 MBO
using the RASE Methodology

Course	RASE / Rule Formalization Project
Team Members	Aryana Hami, Othman, Samih
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1. Introduction

1.1 Background

The construction industry relies heavily on legal and regulatory documents. These regulations are typically written for human interpretation, making them difficult to process automatically using computer systems.

To address this challenge, the RASE methodology provides a structured approach for converting legal text into machine-readable representations. RASE separates legal requirements into four categories:

Category	Meaning
R	Requirement
A	Applicability
S	Selection
E	Exception

This structure enables legal requirements to be transformed into formal rules that can later be evaluated by software systems.

1.2 Project Objective

The objective of this project is to manually analyse §47 of the Musterbauordnung (MBO) using the RASE methodology and produce a correct human-generated reference model.

This reference model will later be used to evaluate whether Large Language Models (LLMs) can reproduce the same annotation automatically.

2. Selected Clause

2.1 Original Regulation

Insert the original German text of §47.

2.2 Why We Selected §47

This clause was selected because it contains:

- measurable requirements
- multiple applications
- conditional rules
- legal exceptions
- measurement definitions

This makes it an ideal candidate for RASE analysis.

3. Initial Legal Analysis

Sentence 1

"Aufenthaltsräume müssen eine lichte Raumhöhe von mindestens 2,40 m haben."

Analysis

Application:

- Aufenthaltsräume

Requirement:

- lichte Raumhöhe ≥ 2.40 m

Sentence 2

"Aufenthaltsräume im Dachraum ..."

Analysis

Application:

- Aufenthaltsräume im Dachraum

Requirements:

- lichte Raumhöhe ≥ 2.20 m
- over at least 50% of net floor area

Sentence 3

"Die Sätze 1 und 2 gelten nicht ..."

Analysis

Exception:

- Wohngebäude GK1
- Wohngebäude GK2

4. RASE Classification

Element	Type
Aufenthaltsräume	Application
Dachraum	Application
2.40 m	Requirement
2.20 m	Requirement
50% Nettofläche	Requirement
GK1	Exception
GK2	Exception

5. Interpretation Challenges

This section documents the key semantic challenges encountered during the RASE annotation process.

Challenge 1 — Is Sentence 2 a Requirement or an Exception?

Initially Sentence 2 appeared to be another independent requirement.

However, after reviewing the clause structure and expert feedback, it became clear that it functions as a special rule for attic rooms and therefore overrides the general rule from Sentence 1.

Challenge 2 — The 1.50 m Statement

"Raumteile mit einer lichten Raumhöhe bis zu 1,50m bleiben außer Betracht."

Initially this statement was interpreted as an exception. However, it was later determined that the sentence defines a measurement methodology rather than a legal requirement. The sentence therefore should not be represented as an independent RASE element.

6. Expert Feedback

Include screenshots from Nick.

Main Feedback Point 1

The title *Aufenthaltsräume* is itself an Application.

Main Feedback Point 2

Sentence 2 should be treated as a special exception section.

Main Feedback Point 3

The 1.50 m statement should not be marked up. It defines area calculation rules.

7. Final RASE Model

ID	Type	Content
A1	Application	Aufenthaltsräume
A2	Application	Dachraum
R1	Requirement	Height $\geq$ 2.40 m
R2	Requirement	Height $\geq$ 2.20 m
R3	Requirement	$\geq$ 50% Net Area
E1	Exception	GK1
E2	Exception	GK2

8. Machine-Readable Interpretation

Rule R1

```
IF
  Room = Aufenthaltsraum
  AND Room != Dachraum
  AND BuildingClass NOT IN {1, 2}

THEN
  Height >= 2.40m
```

Rule R2

```
IF
  Room = Aufenthaltsraum
  AND Room = Dachraum
  AND BuildingClass NOT IN {1, 2}

THEN
  AreaWithHeight >= 2.20m
  >= 50% NetArea
```

9. Human vs Machine Interpretation

We additionally generated machine-readable representations of the clause using AI-assisted analysis. The generated outputs were compared with the manual RASE model.

The comparison showed that AI can correctly identify many RASE elements but may struggle with semantic distinctions such as measurement definitions and special legal exceptions. This highlights the importance of human validation.

10. Reflection

The project provided practical experience in legal rule formalization and demonstrated the challenges of translating natural language regulations into structured representations.

One of the main lessons learned was that legal interpretation requires understanding the meaning and context of a clause rather than relying solely on keywords.

The final RASE model therefore combines both structured methodology and expert interpretation.

11. Preparation for Task 2

The final annotation produced in Task 1 will serve as the ground-truth reference for the next phase of the project.

Task 2 will investigate whether LLMs such as ChatGPT, Claude, Gemini, and Copilot can reproduce the same RASE annotation automatically.

The quality of the generated annotations will be compared against the human-generated reference model developed in this report.